Midnight Buildathon · zero-knowledge fairness

Lootseal

Prove the odds were real. Spoil no surprise.

Loot boxes and mystery mints where the drop table is proven fair, the draw can't be rigged, nothing can be sniped, and your item is delivered to you — encrypted — inside the same proof.

"1% legendary." Says who?

Unverifiable odds

Games and NFT mints publish rates nobody can check. Players widely suspect they're shaded.

Insider sniping

Whoever can see the table — staff, bots watching a public reveal — grabs the rares first. Late buyers get worse odds than advertised.

Regulation is closing in

China and South Korea mandate probability disclosure; Apple and Google require it; Belgium treats paid loot boxes as gambling.

A multi-billion-dollar industry runs on **trust me**.

Why transparency alone fails

Publish the whole table?

- Spoils the surprise — players know exactly what's left.
- Hands bots a map: buy only when the legendaries remain.
- "The rares are gone, stop buying" games.

Publish nothing?

- Nothing to check. Odds are a marketing claim.
- Commit-reveal after the sale proves totals, but not that each draw was fair or that nobody peeked.

We need **selective disclosure**: prove the odds, hide the table. That's exactly what Midnight is for.

How Lootseal works

1 · Commit

Publisher commits the hidden table and proves in ZK that it matches the advertised odds. Bond posted.

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2 · Buy

Buyer pays NIGHT and picks their own seed. The box is sealed in the queue.

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3 · Open

A circuit draws from the hidden remaining supply with the committed seed + buyer seed, and encrypts the item to the buyer.

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4 · Hold & trade

Prove "top 1%" without revealing the item. Gift privately. Sell through escrow.

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5 · Audit

After the sale the table is opened and anyone re-checks it on chain.

Late box? Refunded.

Boxes open FIFO within a window — or the buyer claims price + penalty from the bond.

Nothing to snipe

Remaining supply is a salted commitment, re-committed every open. Every box carries exactly the advertised odds.

Private forever

The master seed is never revealed, so nobody can later publish who got what.

The guarantees, in circuits

Guarantee	Circuit	Proven in zero knowledge
Odds are real	`commitTable`	Σ hidden counts per tier = advertised tier totals
Draw can't be steered	`openBox`	seed opens its commitment · roll = H(seed, buyerSeed, k) · checked reduction & pick
No withholding	`openBox` · `claimRefund`	strict FIFO · deadline · bonded penalty · drop marked DEFAULTED
Only the buyer learns the item	`openBox`	Jubjub Diffie-Hellman ciphertext bound to the item commitment
Rarity without disclosure	`proveRarityTier`	owner key opens the commitment · tier ≤ claim
Fair exchange	`settleListing`	delivers to the escrowed buyer keys *and* pays the seller, atomically
Public audit	`finalReveal` · `verifyOdds`	commitments opened · re-checked on chain by anyone

The product

Drop page

Proven odds, the catalog, a live queue, and a box that shakes, cracks and flips to your item when the proof lands.

Vault

Your items, decrypted locally from encrypted deliveries. Prove rarity, gift with a Lootseal ID, list on the bazaar, back up your keys.

Bazaar

Buy sealed items whose rarity is proven in ZK. Escrowed until the seller delivers — or refunded.

Studio

Design a drop with a live odds preview and launch it in guided steps. Run the queue, claim proceeds, reveal.

Audit & Verify

Every guarantee checked from public data. Rarity badges anyone can verify — no wallet needed.

Fulfiller

Opens boxes automatically within a block or two. Stateless: replays the hidden supply from chain data.

Built on Midnight's dual ledger

Public ledger

- catalog, tier odds, price, deadlines
- table / supply / seed commitments
- request queue · item commitments · ciphertexts
- badges · listings · NIGHT escrow & bond

Private state (provers only)

- per-item counts · remaining supply · master seed
- owner & encryption keys · decrypted items
- encrypted in the browser (IndexedDB, passphrase)

Witness-checked work

Roll reduction and item pick are computed off chain and asserted in circuit — a lying witness can only fail. Secret indices never touch the ledger: vectors are read whole and selected with `fold`/`map`.

Engineering

15 + 1 provable circuits, 15 pure circuits shared with the browser. Staged deploy installs verifier keys via signed maintenance updates to fit block limits.

Tested like it matters

68

simulator tests — every circuit, a rejection test for every reachable assert, lying witnesses

14

devnet end-to-end tests with real proofs, NIGHT, refunds and escrow

72

web-app unit tests

χ^2

legendary equally likely at every box position (150 drops); a sold-out drop delivers exactly the table

Privacy is tested, not claimed

`assertNotInPublicState` after every phase — launch, sales, opens, badges, transfers, bazaar, reveal: no seed, salt, key, shared secret or item opening ever reaches public state.

Honest threat model

Holds against the publisher

- odds proven before sale · reveal must match
- can't pick, skip or reorder boxes
- late boxes refunded from the bond

Holds against everyone else

- no peeking at the remainder — nothing to snipe
- items and holders stay private
- badges and bazaar sales are sound

Residual we state openly

A publisher buying its own boxes through other wallets knows the seed and can bias *its own* pulls (totals still hold, each try costs a box). Fix on the roadmap: **threshold seed keepers** — an independent auditor co-commits the seed, and every open needs both.

Market & roadmap

Who pays

Gacha & mobile studios facing disclosure law · web3 games · NFT launchpads · trading-card platforms.

Model

1–2% per opened box or a flat fee per drop · "Lootseal Verified" badge and audit page · self-hosted fulfiller tier.

Adoption

TypeScript SDK + fulfiller today → Unity / Godot plugins → launchpad integrations.

Next

Threshold seed keepers · shielded purchases · provable pity timers · multi-box packs in one proof · anonymous holder badges (Merkle + nullifiers) · cross-drop publisher reputation.

Lootseal — the fairness layer for every surprise.